A Tale of Five Decoders.

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November 2, 2025

Introduction

Today:

- Recap, and were we stood before the summer.
- Simulations Results.
- Where we heading.

Map

F

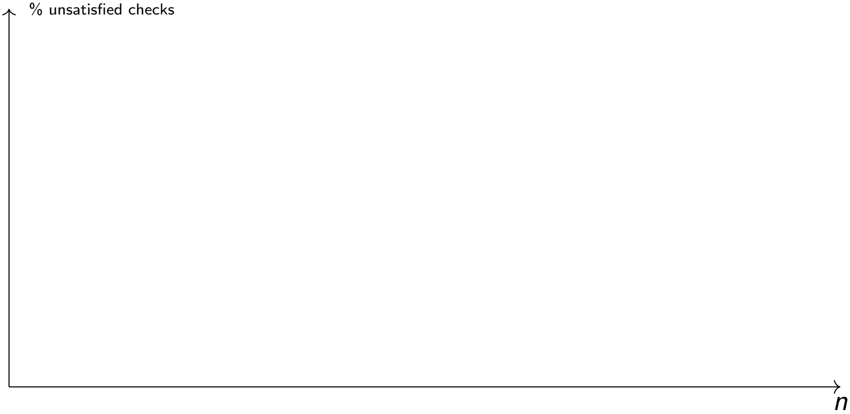
Perskil and Swift Ru

Dennis at el?

Is the Toric a MEM?

Coming with $O(1)$-Depthfor 'good' codes.

How to Read



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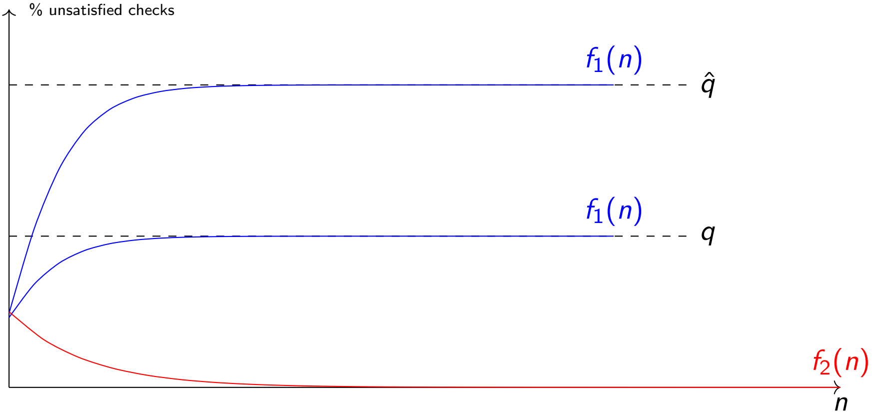
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

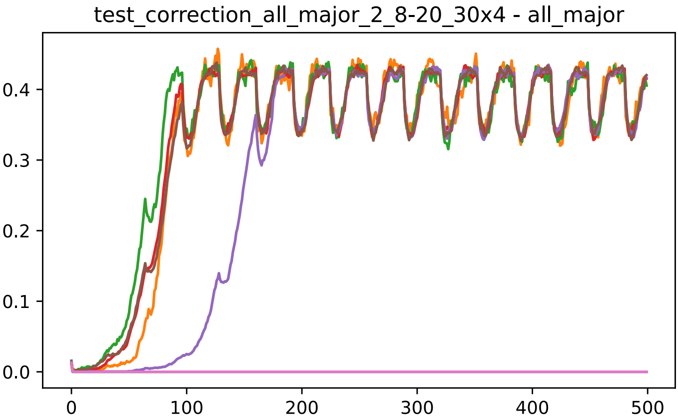
Ideal Result.



colors

In red, No accumulation of errors. In blue decoding in the presence of noise.

Majoritiy



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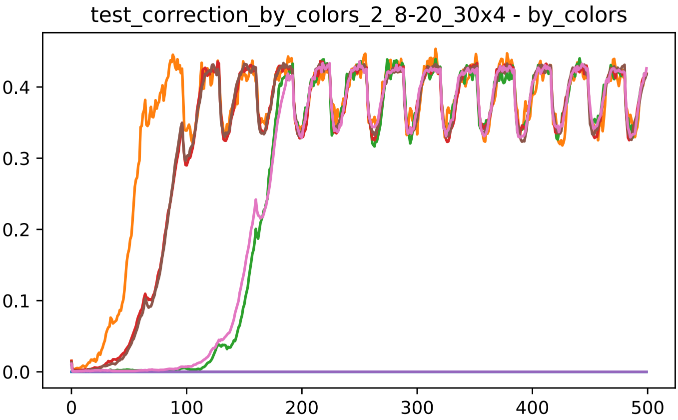
Colors



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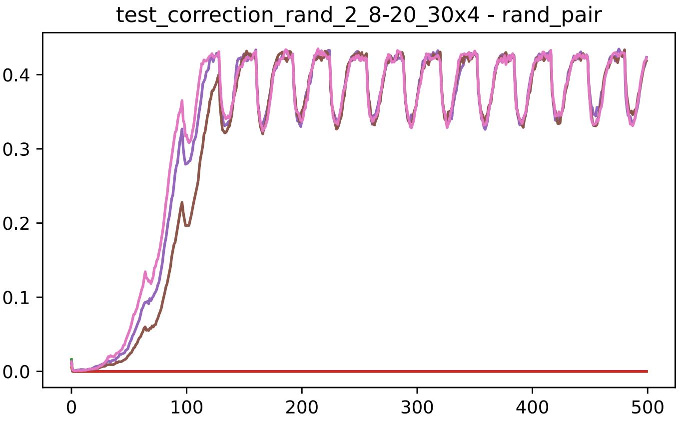
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

Picking Random



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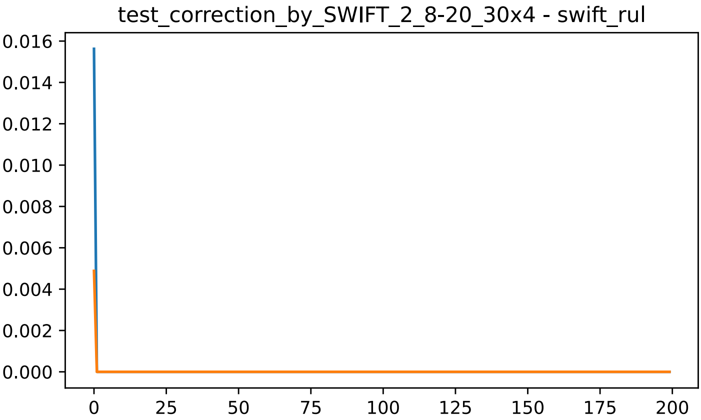
error accumulation :False

SWIFT



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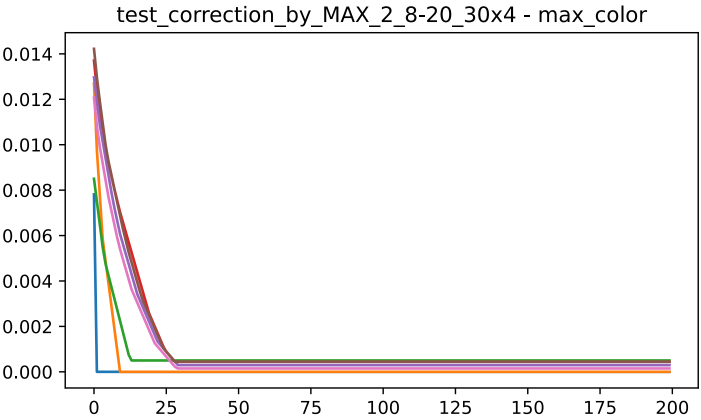
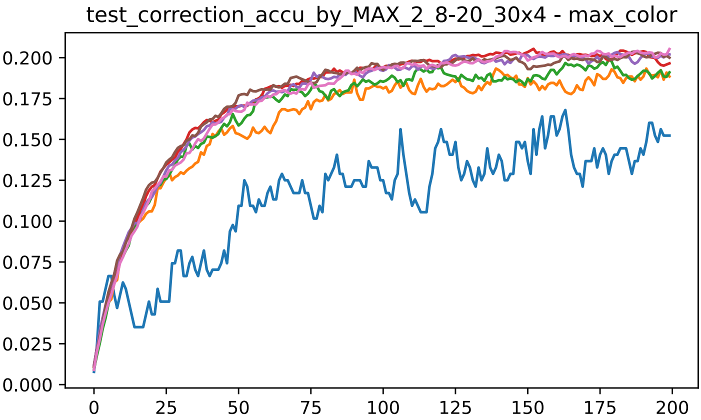
error rates :[0.001]
error accumulation :False

Max Color



side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

Max Color



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error accumulation :True